Causality and Study Design

Yongjin Park

25 January 2026

Course logistics: project proposal

- Project proposal (Thursday)
- Written proposal (February 20)

Roadmap of STAT/BIOF/GSAT540 2026

data generation followed by *inference* or validation

3. Formulating biological questions
4. Hypothesis testing
5. RNA-seq data
6. Inference focusing on RNA-seq
7. **Experimental design** (Today)
8. Single-cell genomics
9. Matrix factorization
10. Genome-wide association studies
11. Multiple hypothesis testing

Learning objectives: Study (analysis) design

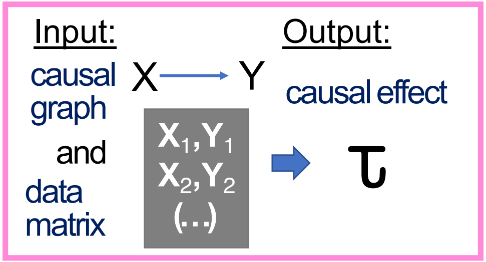
- It would be great if we could design experiments
- In reality, many studies are already set up
- We need to do study design for our analysis

Today's lecture

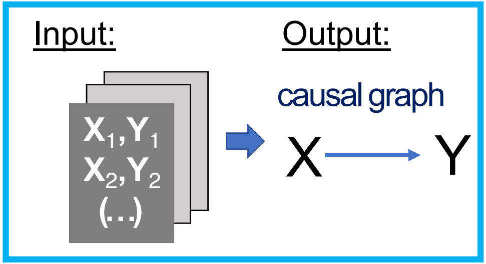
1. Introduction to causal inference
2. How to incorporate causal inference in RNA-seq
3. Removing Unwanted Variation in RNA-seq

Background: What is causal inference?

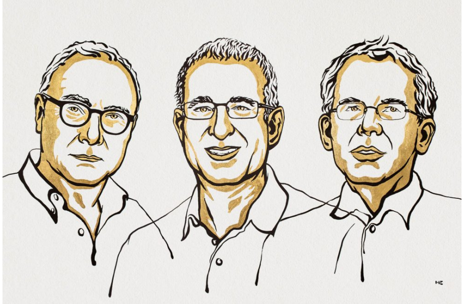
Causal Effect Inference



Causal structure discovery



The Nobel Prize 2021 in Economic Sciences



David Card, $\underbrace{\text{Joshua Angrist}}_{\text{instrumental variable}}$, $\underbrace{\text{Guido Imbens}}_{\text{matching, propensity}}$

"For the methodological contributions to the analysis of causal relationships"

https://www.nobelprize.org/prizes/economic-sciences/2021

The goal of causal effect inference

1. Instead of **"seeing"** what happened (data analysis), we want to measure **the effect of "doing"** an experiment (intervention).

The goal of causal effect inference

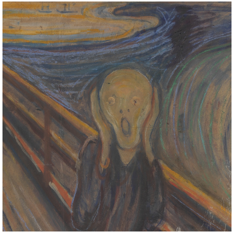
1. Instead of **"seeing"** what happened (data analysis), we want to measure **the effect of "doing"** an experiment (intervention).
2. Because correlation is not causation.

The goal of causal effect inference

1. Instead of **"seeing"** what happened (data analysis), we want to measure **the effect of "doing"** an experiment (intervention).
2. Because correlation is not causation.
3. If it is not causation, then what is it?

Common misconception - correlation implies causation

Henry Niles also wrote, "*To contrast 'causation' and 'correlation' is unwarranted* ***because causation is simply perfect correlation***"



Pearl, *The Book of Why*, p.78

- How do we measure causal effects? Will any experiment do?

A working example: Why correlation is not causation

- U_i: The age at which a person i enrolled in this study.
- X_i: An indicator variable for whether a physician decided to administer this new preventative drug ($X_i = 1$) or not ($X_i = 0$).
- Y_i: Biomarker protein concentration (e.g., Amyloid-beta protein disrupts normal brain functions).

A hypothetical data generating scheme

- A person i's age U_i.

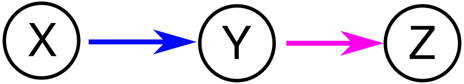
1. Prescribe the preventative drug based on the subject's age U_i (as they might need one):

$$P(X_i = 1|U_i) = \sigma\left(U_i\delta\right)$$

2. A biomarker protein concentration:

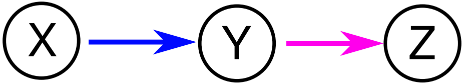
$$\mathbb{E}[Y_i|X,U] = X_i\tau + U_i\beta$$

Recap: Directed Acyclic Graph



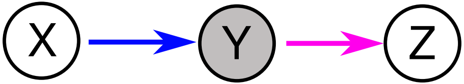
$$p(X, Y, Z) = \underbrace{p(Y|X)}_{\text{first edge}} \quad \underbrace{p(Z|Y)}_{\text{second edge}} \quad \underbrace{p(X)}_{\text{first node}}$$

DAG: Causal path/trail and reachability



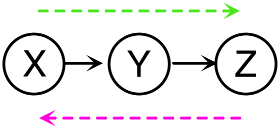
- There is a path between X and Y
- Also between Y and Z
- Also from X to Z

DAG: Conditioning (and/or adjustment)



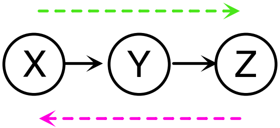
- A shaded node = conditioning like $P(Z|Y = y^\star)$ and $P(Y = y^\star|X)$
- There is a path between X and Y
- Also between Y and Z
- **But** no path from X to Z

d-separation: testing conditional independence (flow vs. no flow)

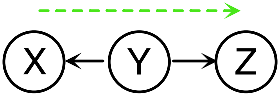


Flow: The outcome of Z depends on the effect of X

d-separation: testing conditional independence (flow vs. no flow)

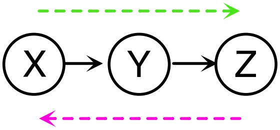


Flow: The outcome of Z depends on the effect of X

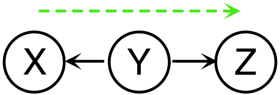


Flow: The outcome of Z depends on the effect of Y; so does the outcome of X on that of Y

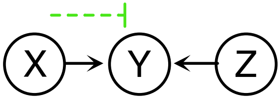
d-separation: testing conditional independence (flow vs. no flow)



Flow: The outcome of Z depends on the effect of X

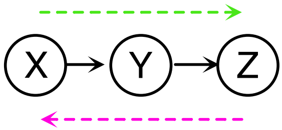


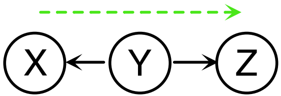
Flow: The outcome of Z depends on the effect of Y; so does the outcome of X on that of Y

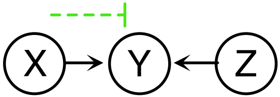


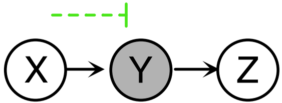
No Flow: The outcome of Z only affects Y; so does X

d-separation: testing conditional independence (flow vs. no flow)



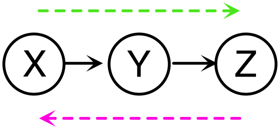


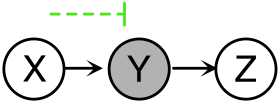




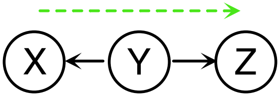
No Flow: The outcome of Z solely depends on $Y = y^*$, not X

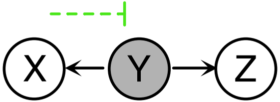
d-separation: testing conditional independence (flow vs. no flow)



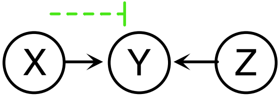


No Flow: The outcome of Z solely depends on $Y = y^*$, not X

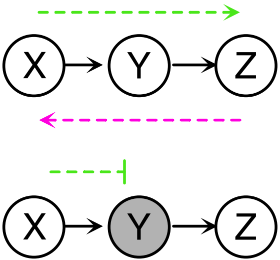




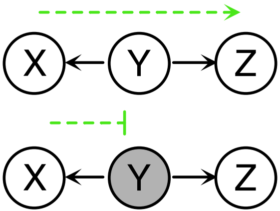
No Flow: The outcome of Z solely depends on $Y = y^*$, not X



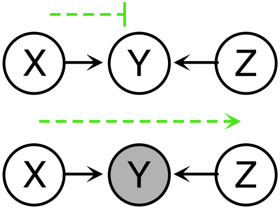
d-separation: testing conditional independence (flow vs. no flow)



No Flow: The outcome of Z solely depends on $Y = y^*$, not X

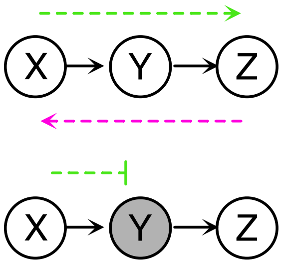


No Flow: The outcome of Z solely depends on $Y = y^*$, not X

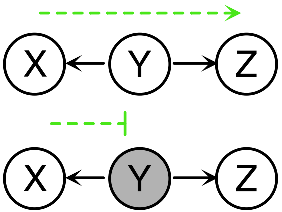


Flow: The observed outcome $Y = y^*$ could stem from X or/and Z

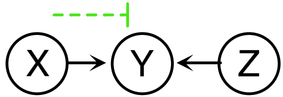
d-separation: testing conditional independence (flow vs. no flow)



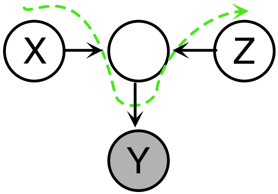
No Flow: The outcome of Z solely depends on $Y = y^*$, not X



No Flow: The outcome of Z solely depends on $Y = y^*$, not X



Flow: deeper v-structure



Going back to the same working example

```r
set.seed(1)
```

If there were no confounding:

```r
yy.unc <- xx * .tau + rnorm(n) * .eps
```

- The variable U confounds X and Y.
- How can we represent it graphically?

Going back to the same working example

```r
set.seed(1)
```

If there were no confounding:

```r
yy.unc <- xx * .tau + rnorm(n) * .eps
```

- The variable U confounds X and Y.
- How can we represent it graphically?

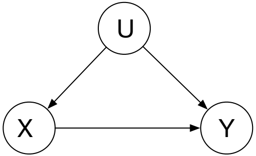
Going back to the same working example

```r
set.seed(1)
```

If there were no confounding:

```r
yy.unc <- xx * .tau + rnorm(n) * .eps
```

- The variable U confounds X and Y.
- How can we represent it graphically?



1. How can we identify the causal effect, $X \rightarrow Y$?

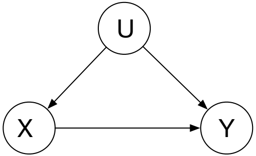
Going back to the same working example

```r
set.seed(1)
```

If there were no confounding:

```r
yy.unc <- xx * .tau + rnorm(n) * .eps
```

- The variable U confounds X and Y.
- How can we represent it graphically?



1. How can we identify the causal effect, $X \to Y$?
2. Why can't we just report the correlation/association between X and Y?

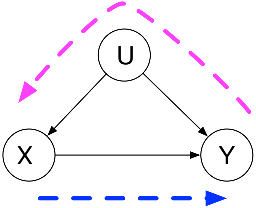
Going back to the same working example

```r
set.seed(1)
```

If there were no confounding:

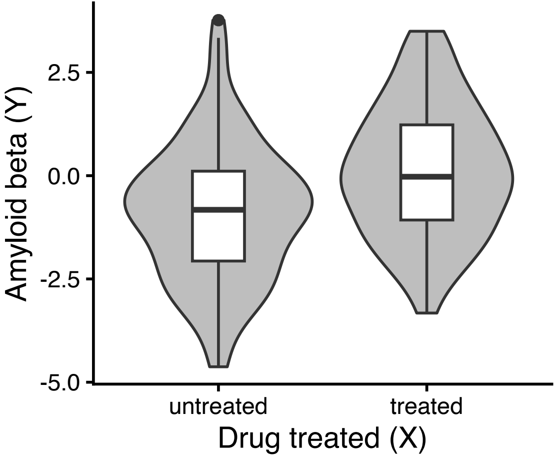
```r
yy.unc <- xx * .tau + rnorm(n) * .eps
```

- The variable U confounds X and Y.
- How can we represent it graphically?



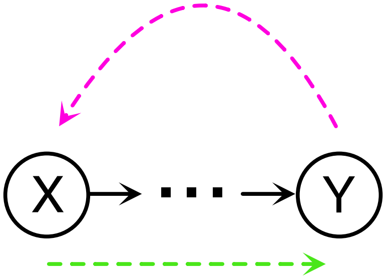
1. How can we identify the causal effect, $X \to Y$?
2. Why can't we just report the correlation/association between X and Y?

Differential biomarker analysis suggests that our drug can exacerbate AD



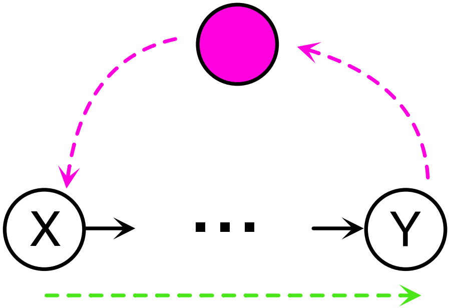
It happened because

Question: Which nodes should be "conditioned" and/or "adjusted" to block a reverse path from Y to X?

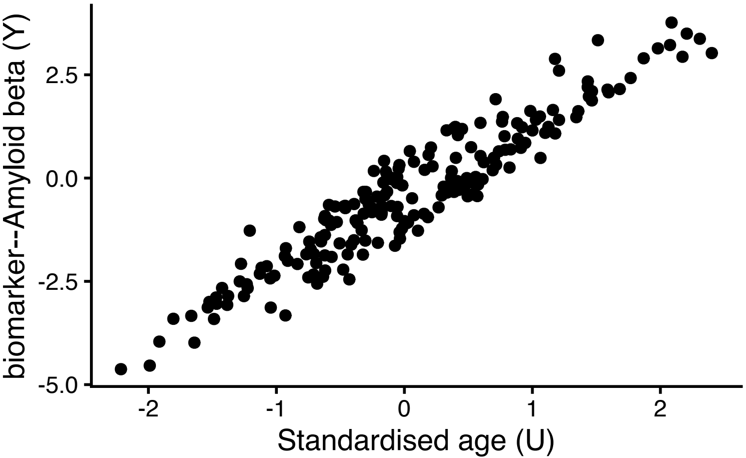


It happened because

Question: Which nodes should be "conditioned" and/or "adjusted" to block a reverse path from Y to X?

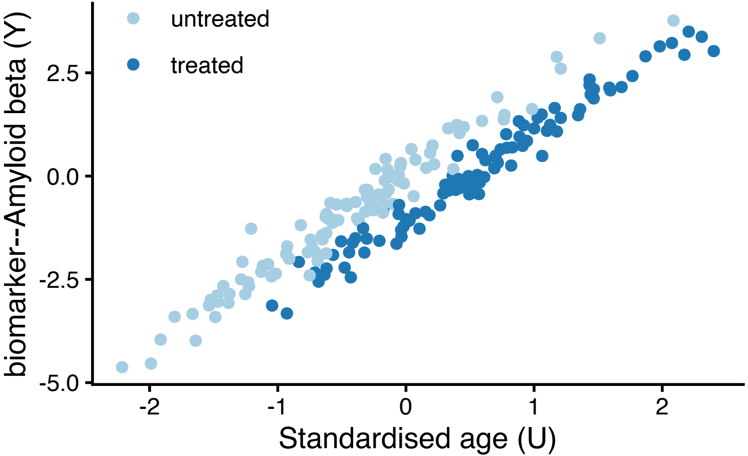


Prevalence of AD increases with age



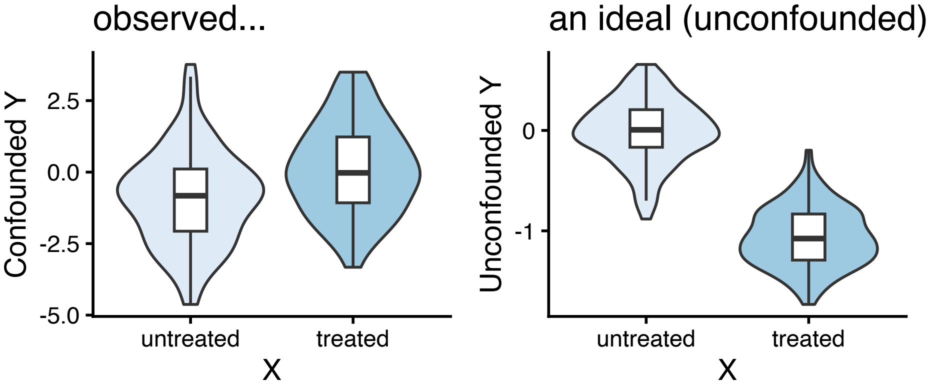
Doctors could have prescribed our drug more frequently to older people because the prevalence of AD tends to increase with age.

Prevalence of AD increases with age



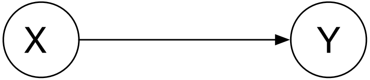
Doctors could have prescribed our drug more frequently to older people because the prevalence of AD tends to increase with age.

Only if we could observe the unconfounded data



What could have been an ideal experimental design?

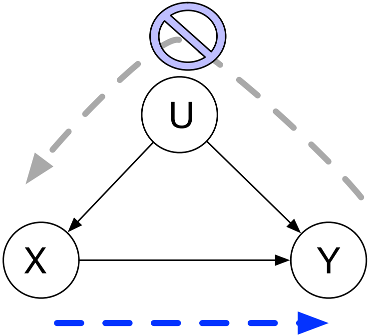
There is no "backdoor" between a putative cause variable (X) *and outcome variable* (Y).



What could have been an ideal experimental design?

There is no "backdoor" between a putative cause variable (X) and outcome variable (Y).

At least...



RCT: an ideal experiment (feat. RA Fisher)

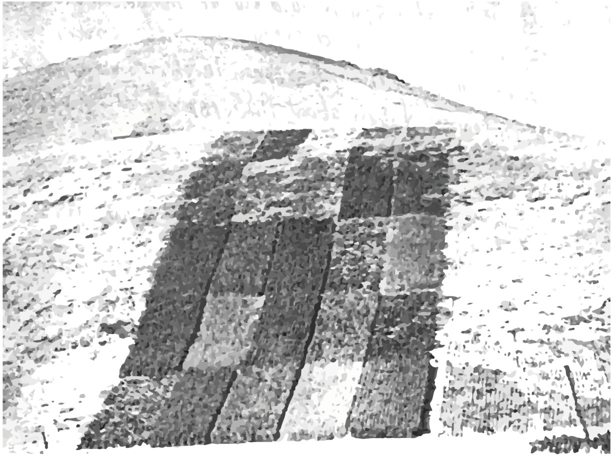
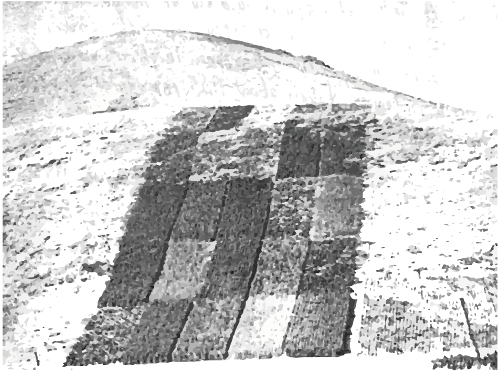


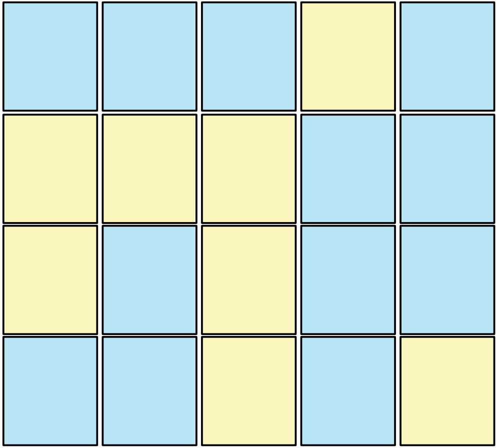
Image source: ‘www.adelaide.edu.au‘

RCT: an ideal experiment (feat. RA Fisher)

Crops grown in the plots

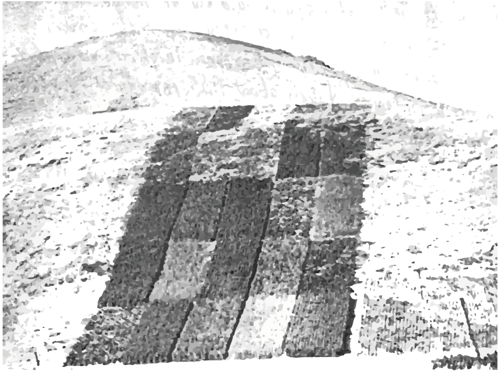


Experimental design

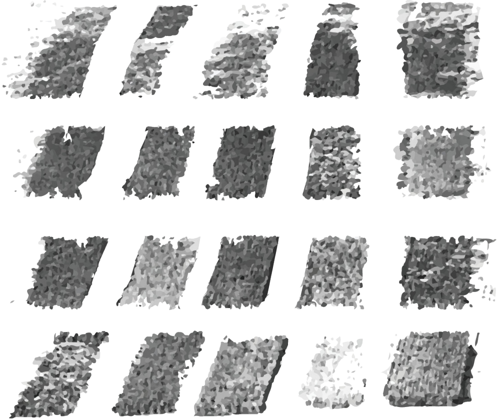


RCT: an ideal experiment (feat. RA Fisher)

Crops grown in the plots

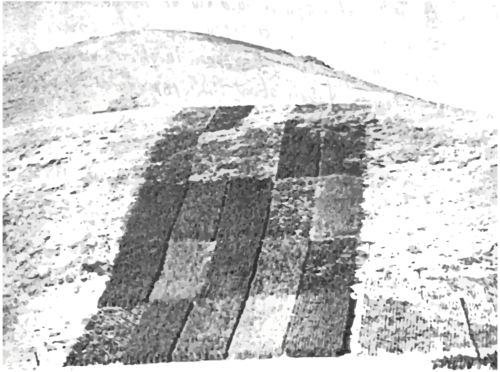


Random assignment of fertilizer

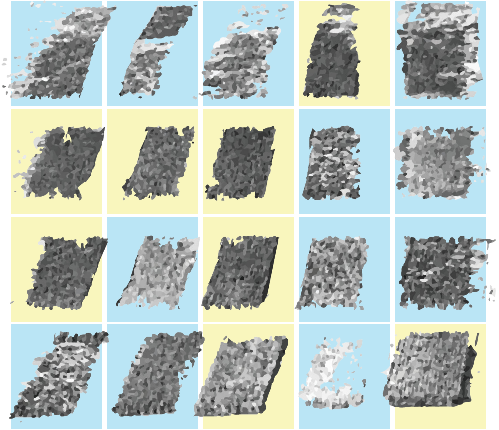


RCT: an ideal experiment (feat. RA Fisher)

Crops grown in the plots

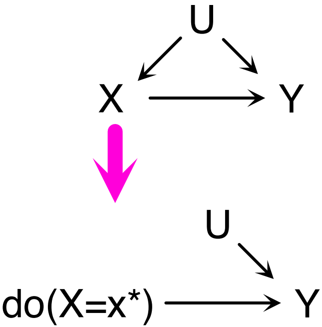


The treated vs. untreated

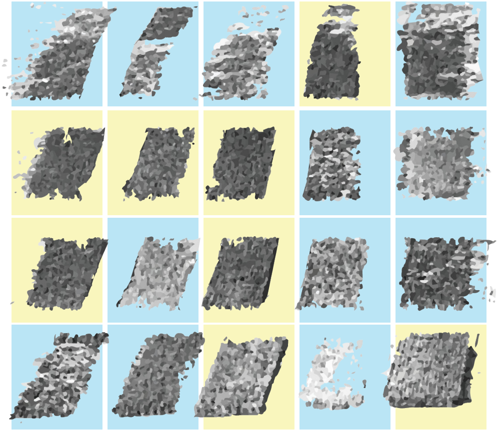


RCT: an ideal experiment (feat. RA Fisher)

Randomised Control Trial $\Longrightarrow$ "doing"

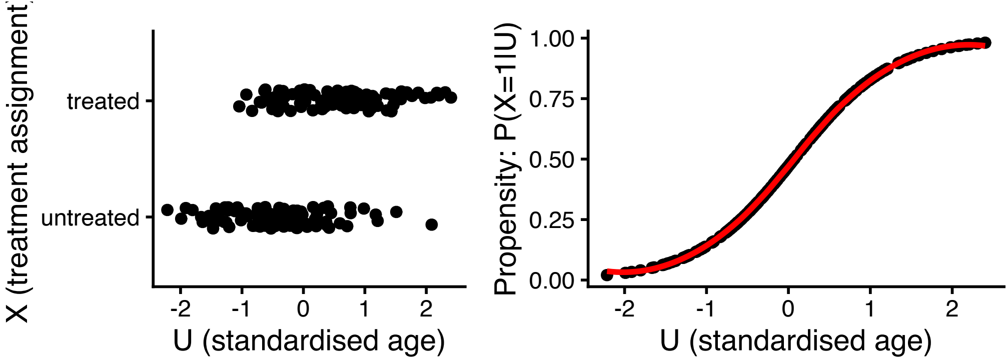


The treated vs. untreated

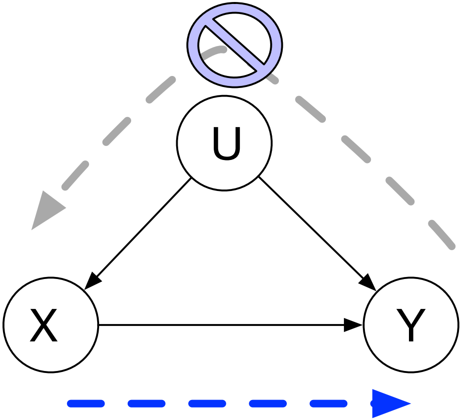


Back to the AD example: Can we save the study?

If there is any systematic bias, we should adjust for it.



Inverse Propensity Weighting reverses confounding




```r
p.xu <-
    glm(x~u, family="binomial") %>%
    predict() %>%
    sigmoid() %>%
    clamp() # avoid strict 0/1

ww <- x / p.xu + (1-x) / (1-p.xu)
```

Propensity: Pr. of assignment $X = 1$:

$$\log \frac{\mathbf{p(X|U)}}{1 - \mathbf{p(X|U)}} \approx \beta_0 + \beta_1 U$$

Inverse Propensity Weighting reverses confounding

$$\hat{Y}_i^{(1)} = \frac{X_i Y_i}{\hat{p}(X_i = 1|U_i)}$$

$$\hat{Y}_i^{(0)} = \frac{(1 - X_i) Y_i}{1 - \hat{p}(X_i = 1|U_i)}$$

```
p.xu <-
    glm(x~u, family="binomial") %>%
    predict() %>%
    sigmoid() %>%
    clamp() # avoid strict 0/1

ww <- x / p.xu + (1-x) / (1-p.xu)
```

Propensity: Pr. of assignment $X = 1$:

$$\log \frac{\mathbf{p(X|U)}}{1 - \mathbf{p(X|U)}} \approx \beta_0 + \beta_1 U$$

Inverse Propensity Weighting reverses confounding

$$\hat{Y}_i^{(1)} = \frac{X_i Y_i}{\hat{p}(X_i = 1|U_i)}$$

$$\hat{Y}_i^{(0)} = \frac{(1 - X_i) Y_i}{1 - \hat{p}(X_i = 1|U_i)}$$

equivalently give weights for ∀ i

$$W_i \propto \begin{cases} 1/\hat{p}(X_i = 1|U_i) & X_i = 1 \\ 1/\hat{p}(X_i = 0|U_i) & X_i = 0 \end{cases}$$

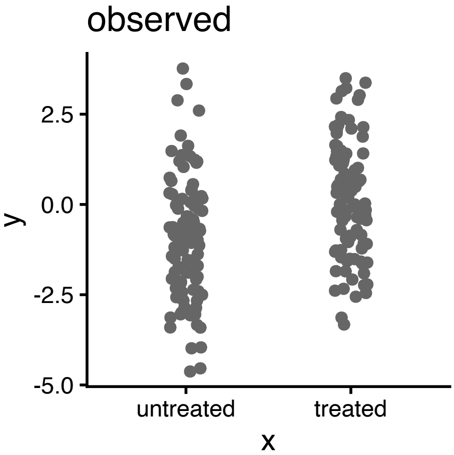
```
p.xu <-
    glm(x~u, family="binomial") %>%
    predict() %>%
    sigmoid() %>%
    clamp() # avoid strict 0/1

ww <- x / p.xu + (1-x) / (1-p.xu)
```

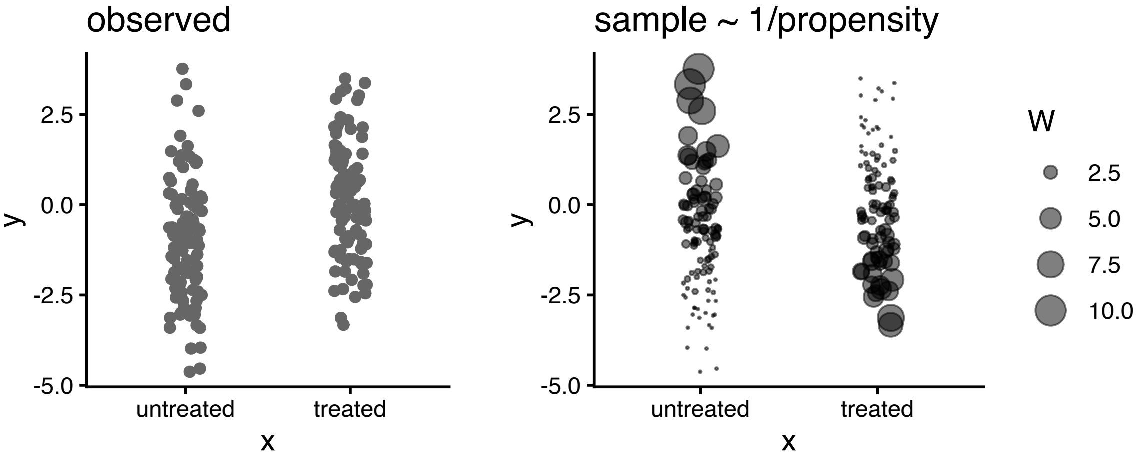
Propensity: Pr. of assignment $X = 1$:

$$\log \frac{\mathbf{p(X|U)}}{1 - \mathbf{p(X|U)}} \approx \beta_0 + \beta_1 U$$

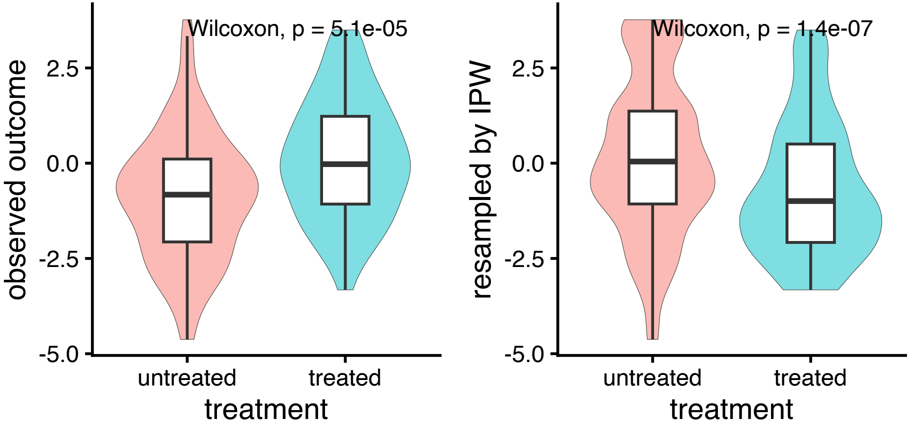
Take samples inversely proportional to propensity



Take samples inversely proportional to propensity



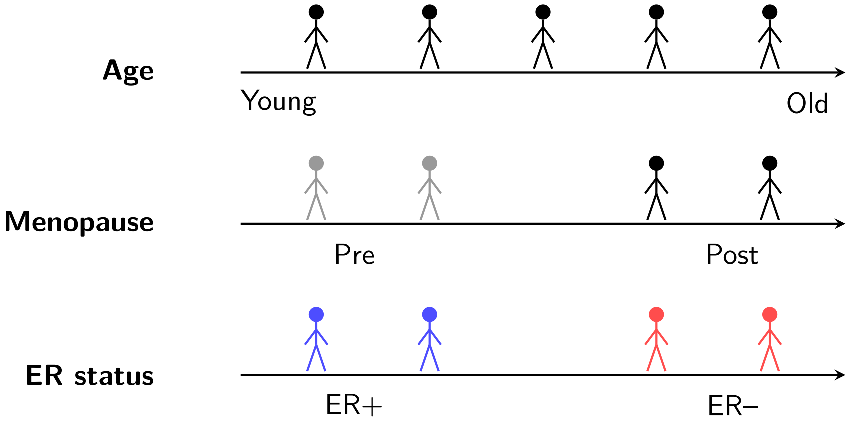
If we bootstrap (sample with replacement)...



Today's lecture

1. Introduction to causal inference
2. How to incorporate causal inference in RNA-seq
3. Removing Unwanted Variation in RNA-seq

TCGA BRCA example: three axes of variation



Key question: multiple sources of variation

- Can we identify multiple independent disease mechanisms?
- How do they manifest in observed transcriptional profiles?

1. **ER status**: Tumour-intrinsic differences (ER+ vs ER-)
2. **Age**: age effects in general population
3. **Menopause**: pre- vs. post-menopause

They are all correlated... but not necessarily the same

- Post-menopausal but young (e.g., surgical menopause)
- Pre-menopausal but older (late menopause)

TCGA BRCA data from `recount3`

We will use The Cancer Genome Atlas (TCGA) breast cancer data

```
data.dir <- "../data/recount3"
dir.create(data.dir, recursive = TRUE, showWarnings = FALSE)
.file <- file.path(data.dir, "tcga_brca_rse.rds")

if.needed(.file, {
    tcga.proj <- subset(available_projects(),
                        file_source == "tcga")
    brca.info <- tcga.proj[tcga.proj$project == "BRCA", ]
    rse.brca <- create_rse(brca.info)
    saveRDS(rse.brca, .file)
})
rse <- readRDS(.file)
```

After alignment: `SummarizedExperiment`

```
                       samples (columns)
                    +----------------------+
                    |  S1   S2   S3   ...  |
              +-----|----------------------+
genes         | G1  |  10   25   18   ...  |  <- assays (counts)
(rows)        | G2  |   5   12    8   ...  |
              | ... | ...  ...  ...   ...  |
              +-----+----------------------+

rowData: genomic coordinates (chr, start, end, strand)
colData: sample metadata (tumour/normal, clinical info)
```

Here is gene expression count

```r
raw.counts <- assay(rse, "raw_counts")
raw.counts[1:3, 1:3]
```

```
##                    0b1cf6de-f98f-4020-baa7-c4fc518e5f00
## ENSG00000278704.1                                     0
## ENSG00000277400.1                                     0
## ENSG00000274847.1                                     0
##                    20c90e97-e900-488e-af6a-6648d9a8bb01
## ENSG00000278704.1                                     0
## ENSG00000277400.1                                     0
## ENSG00000274847.1                                     0
##                    0a2fa7c7-0810-4d65-b16c-4a1792714802
## ENSG00000278704.1                                     0
## ENSG00000277400.1                                     0
## ENSG00000274847.1                                     0
```

Gene/row annotations

```
rowData(rse) %>% head(3)
```

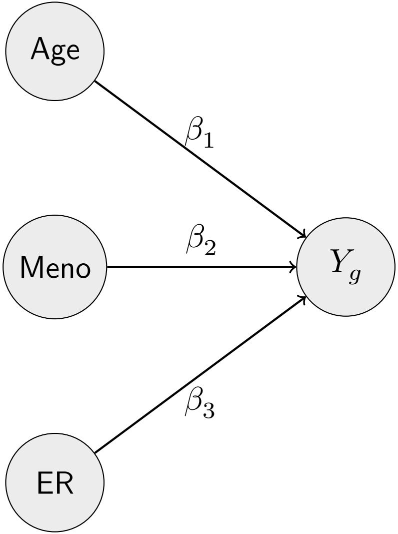
```
## DataFrame with 3 rows and 10 columns
##                       source     type bp_length     phase           gene_id
##                     <factor> <factor> <numeric> <integer>       <character>
## ENSG00000278704.1  ENSEMBL     gene      2237        NA ENSG00000278704.1
## ENSG00000277400.1  ENSEMBL     gene      2179        NA ENSG00000277400.1
## ENSG00000274847.1  ENSEMBL     gene      1599        NA ENSG00000274847.1
##                        gene_type   gene_name       level havana_gene
##                      <character> <character> <character> <character>
## ENSG00000278704.1 protein_coding  BX004987.1           3          NA
## ENSG00000277400.1 protein_coding  AC145212.2           3          NA
## ENSG00000274847.1 protein_coding  AC145212.1           3          NA
##                           tag
##                     <character>
## ENSG00000278704.1          NA
## ENSG00000277400.1          NA
## ENSG00000274847.1          NA
```

Sample/column annotations

```r
colData(rse) %>% as.data.frame() %>% select(1:3) %>% head(3)
```

```
##                                      rail_id
## 0b1cf6de-f98f-4020-baa7-c4fc518e5f00  106628
## 20c90e97-e900-488e-af6a-6648d9a8bb01  106636
## 0a2fa7c7-0810-4d65-b16c-4a1792714802  106645
##                                                               external_id study
## 0b1cf6de-f98f-4020-baa7-c4fc518e5f00 0b1cf6de-f98f-4020-baa7-c4fc518e5f00  BRCA
## 20c90e97-e900-488e-af6a-6648d9a8bb01 20c90e97-e900-488e-af6a-6648d9a8bb01  BRCA
## 0a2fa7c7-0810-4d65-b16c-4a1792714802 0a2fa7c7-0810-4d65-b16c-4a1792714802  BRCA
```

A probabilistic model on gene expression Y



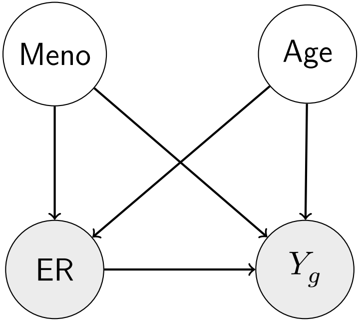
Linear model for gene g:

$$Y_{ig} = \beta_{1g} \cdot X_{i1} + \beta_{2g} \cdot X_{i2} + \beta_{3g} \cdot X_{i3} + \epsilon_{ig}$$

where $\epsilon_{ig} \sim N(0, \sigma_g^2)$

- X_{i1}: Age for individual i
- X_{i2}: Menopause status for individual i
- X_{i3}: ER status for individual i
- $\beta_{1g}, \beta_{2g}, \beta_{3g}$: Coefficients for gene g

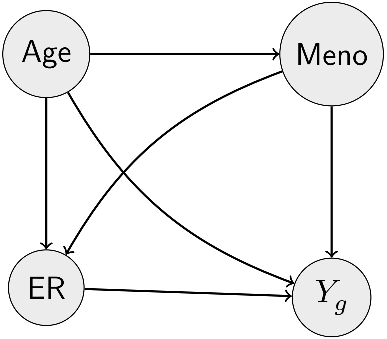
What if Age and Menopause are unobserved?



Goal: Estimate the causal effect of ER on Y

Question: What could go wrong?

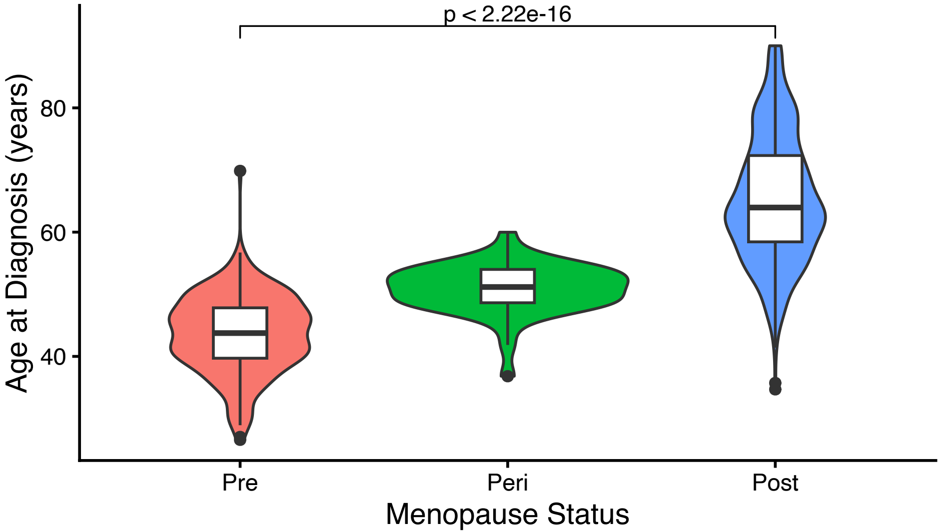
Overall graphical model



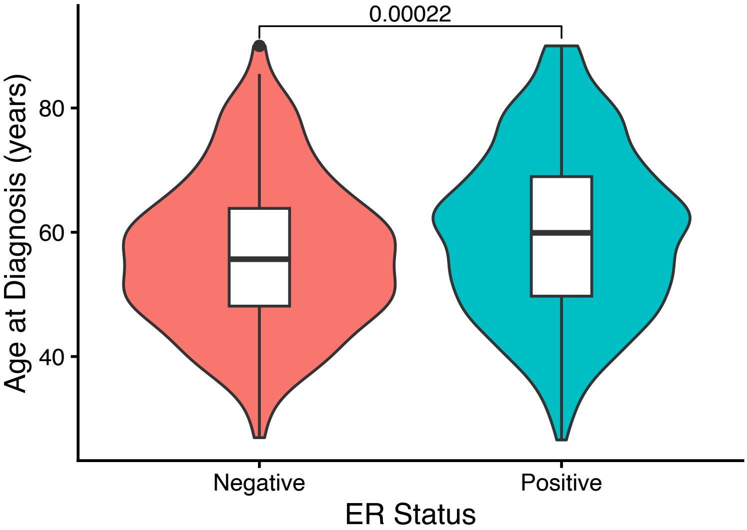
Observed variables:

- Age, Menopause, ER status
- Gene expression Y_g

Age distribution by menopause status



Age distribution by ER status



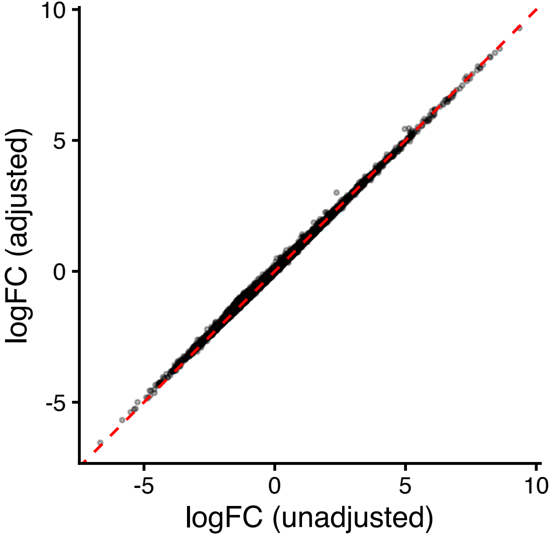
ER effect: with vs. without adjustment

Unadjusted:

$$Y_g \sim \text{ER}$$

Adjusted:

$$Y_g \sim \text{ER} + \text{Age} + \text{Meno}$$



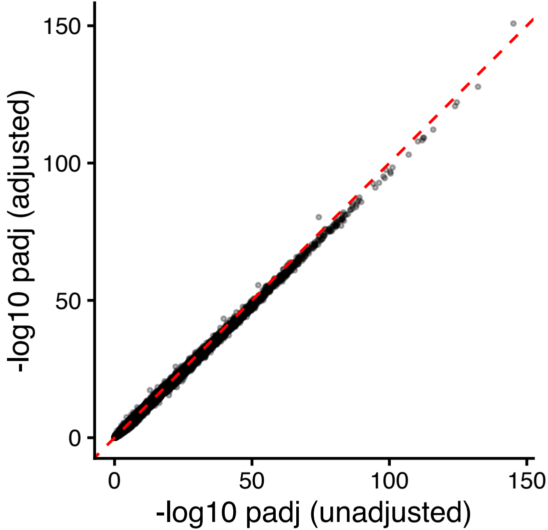
ER effect: with vs. without adjustment

Unadjusted:

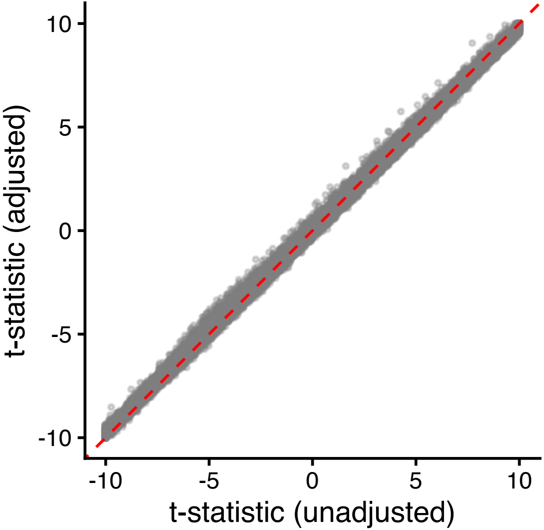
$$Y_g \sim \text{ER}$$

Adjusted:

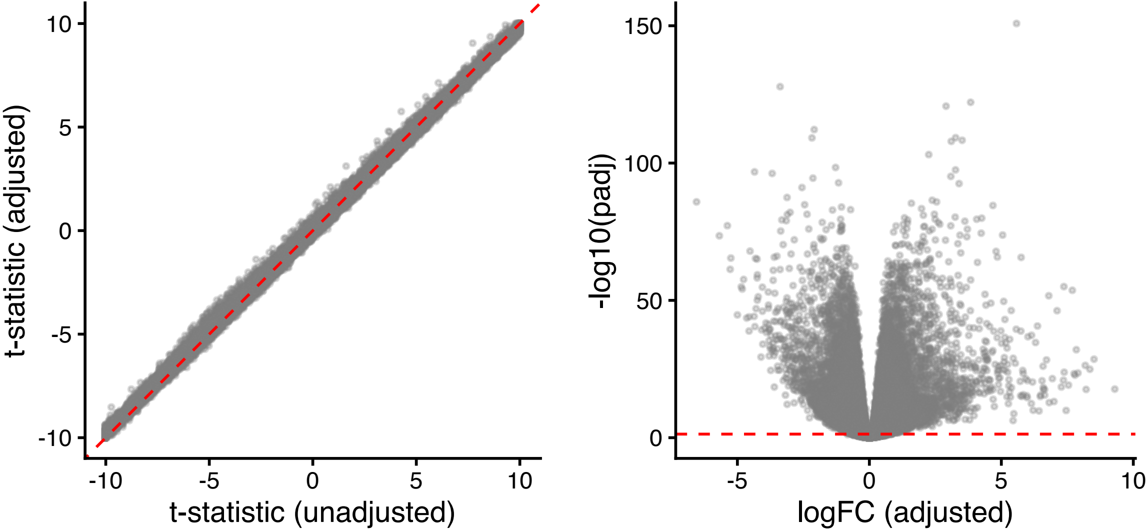
$$Y_g \sim \text{ER} + \text{Age} + \text{Meno}$$



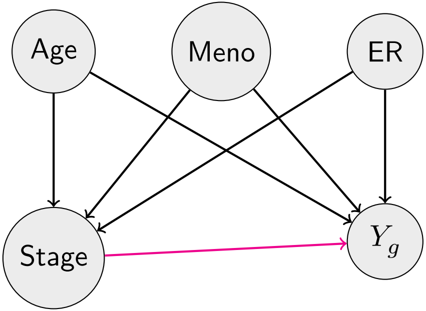
ER effect: t-statistic comparison



ER effect: t-statistic comparison



Confounders of the tumour stage effect



Goal: Estimate the causal effect of tumour stage on Y

Confounders: Age, Meno, ER

- Affect tumour stage at diagnosis
- Also affect gene expression

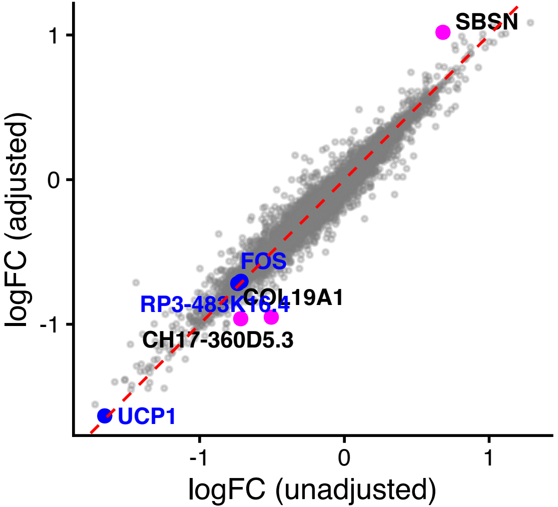
Tumour stage effect: with vs. without adjustment

Unadjusted:

$$Y_g \sim \text{Stage}$$

Adjusted:

$$Y_g \sim \text{Stage} + \text{Age} + \text{Meno} + \text{ER}$$



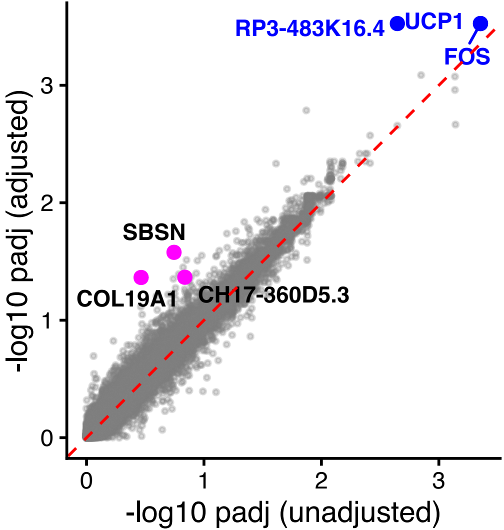
Tumour stage effect: with vs. without adjustment

Unadjusted:

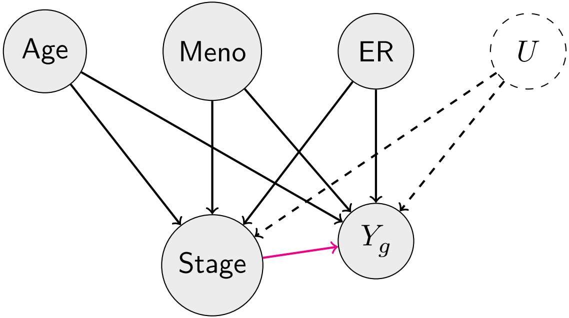
$$Y_g \sim \text{Stage}$$

Adjusted:

$$Y_g \sim \text{Stage} + \text{Age} + \text{Meno} + \text{ER}$$



Confounders: observed and unobserved



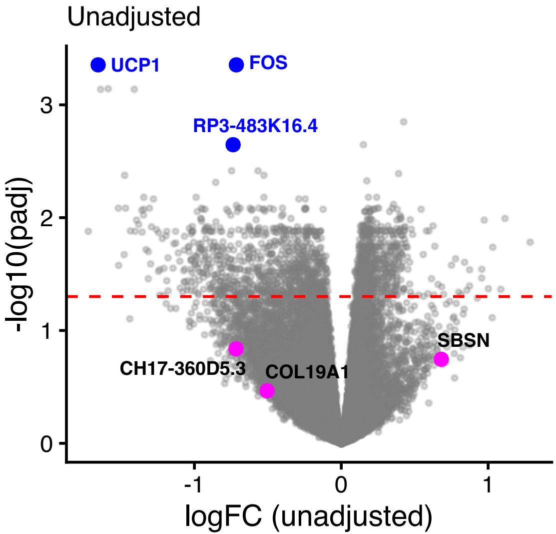
Observed confounders: Age, Meno, ER

- Can be measured and adjusted

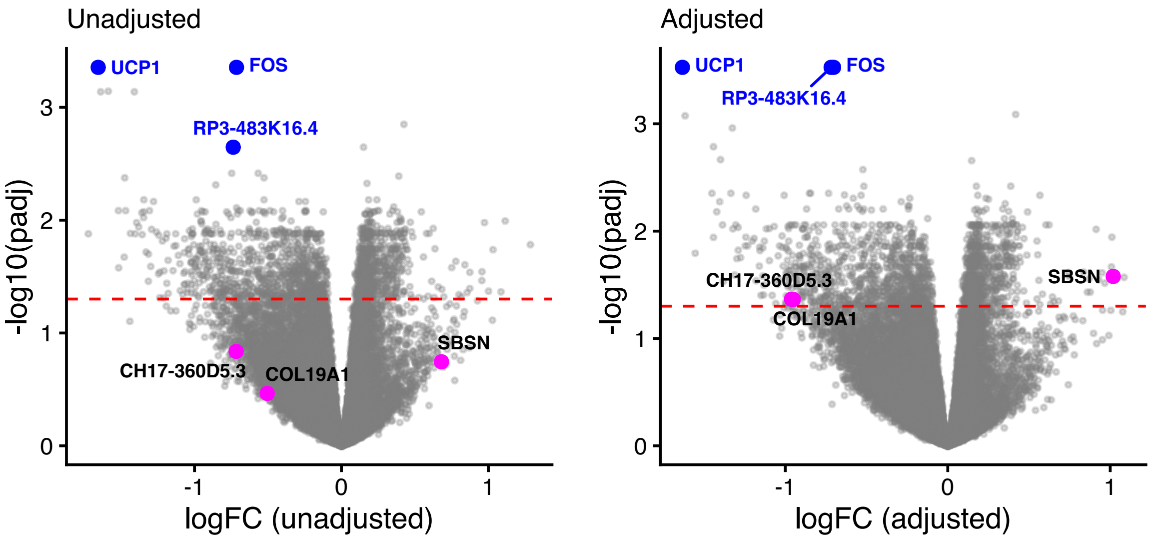
Unobserved confounders: U

- Batch effects, sample quality
- Stromal/immune cell contamination
- Technical artifacts
- Cannot directly measure $\rightarrow$ estimate via PCA/RUV

Tumour effects markedly change with PC adjustment



Tumour effects markedly change with PC adjustment



SBSN promotes tumour endothelial migration and therapy resistance

Expression in Breast Cancer:

- Upregulated in tumour endothelial cells (TEC) vs. normal endothelial cells (Alam et al. 2014)
- Induced by radio- or chemotherapy in breast cancer cell lines
- Elevated in cancer stem-like cells with low-adhesive phenotype

Function:

- Promotes migration and tube formation in TEC
- Activates AKT signaling pathway
- Involved in therapy resistance and anoikis resistance
- Associated with poor prognosis in multiple cancers (Tan et al. 2022)

COL19A1 loss marks basement membrane invasion

Loss during invasion:

- Completely absent in invasive breast carcinomas (Amenta et al. 2003)
- Loss precedes basement membrane invasion
- Occurs earlier than type IV collagen and laminin loss
- Signals ECM remodelling to promote tumour infiltration

Anti-tumour NC1 domain:

- NC1 domain inhibits tumour cell migration and invasion (Oudart et al. 2016)
- Works through integrin v 3 receptor
- Inhibits FAK/PI3K/Akt/mTOR pathway
- Anti-angiogenic properties
- Plasmin cleavage releases protective fragments

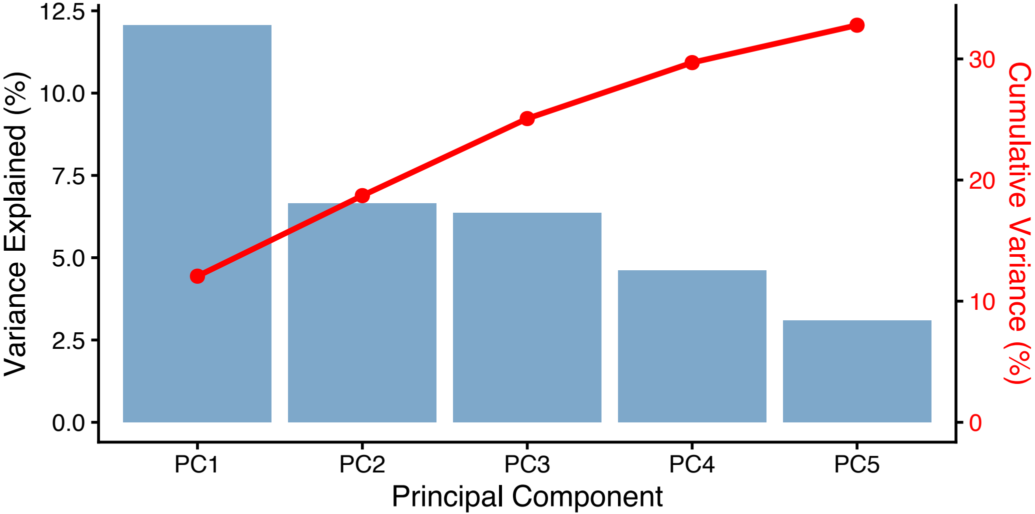
Did we capture all the confounding effects?

What if we have unmeasured confounding variables?

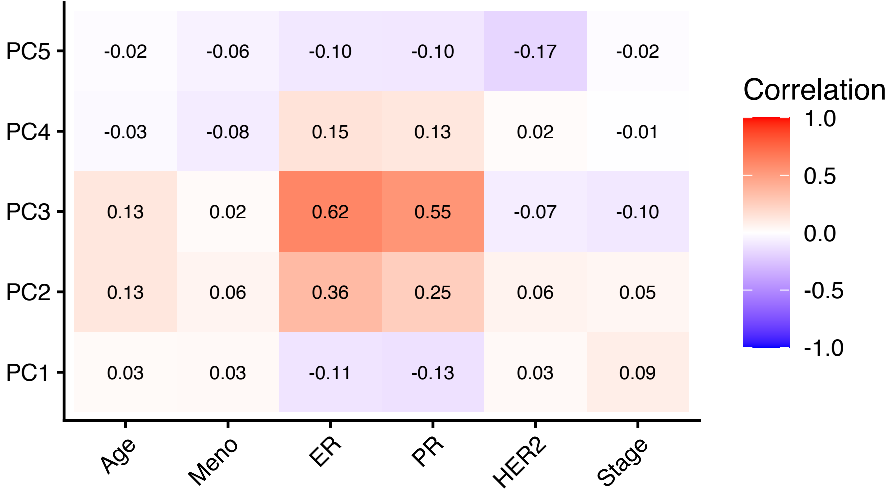
- **A solution** (`lect09`): **Principal Component Analysis** (Hotelling 1933; Jolliffe 1986)

- PCA decomposes total variance of a matrix into independent components.

PCA: variance explained by top components



PCs vs. clinical covariates: correlation heatmap



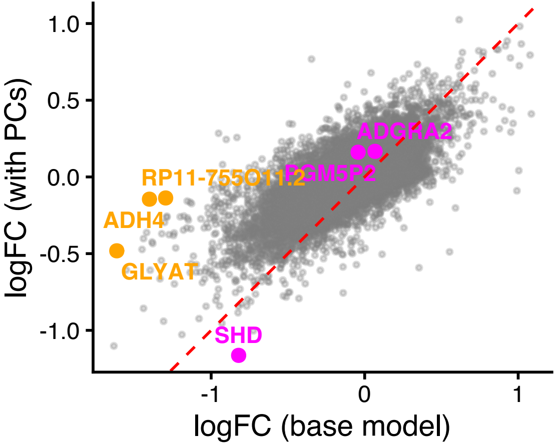
Tumour stage effect: with vs. without PC adjustment

Model 1 (with known covariates):

$$Y_g \sim \text{Stage} + \text{Age} + \text{Meno} + \text{ER}$$

Model 2 (with PCs added):

$$Y_g \sim \text{Stage} + \text{Age} + \text{Meno} + \text{ER} + \text{PC1-5}$$



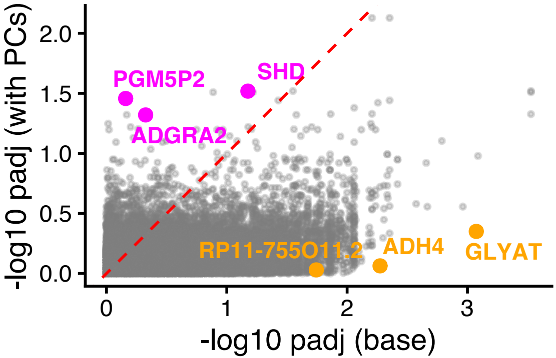
Tumour stage effect: with vs. without PC adjustment

Model 1 (with known covariates):

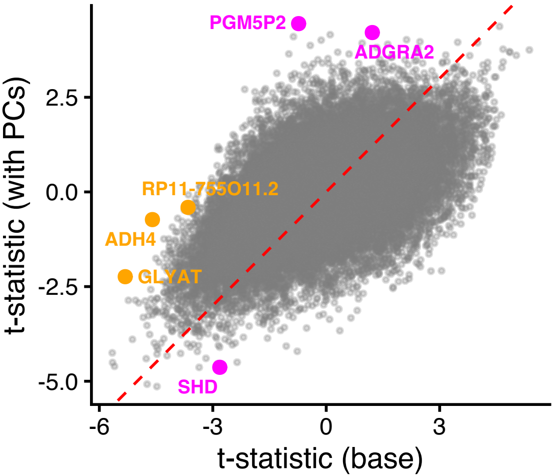
$$Y_g \sim \text{Stage} + \text{Age} + \text{Meno} + \text{ER}$$

Model 2 (with PCs added):

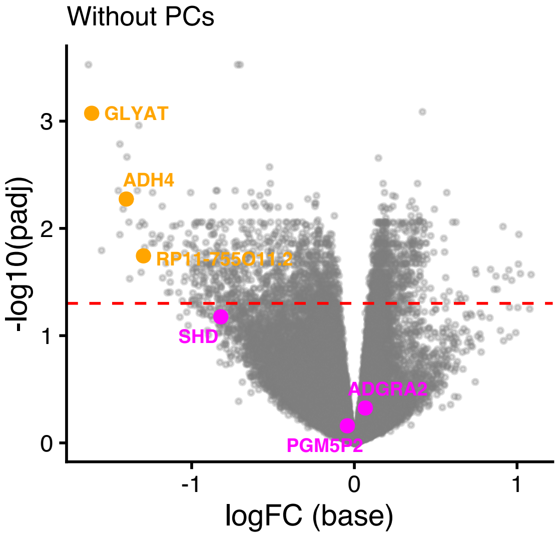
$$Y_g \sim \text{Stage} + \text{Age} + \text{Meno} + \text{ER} + \text{PC1-5}$$



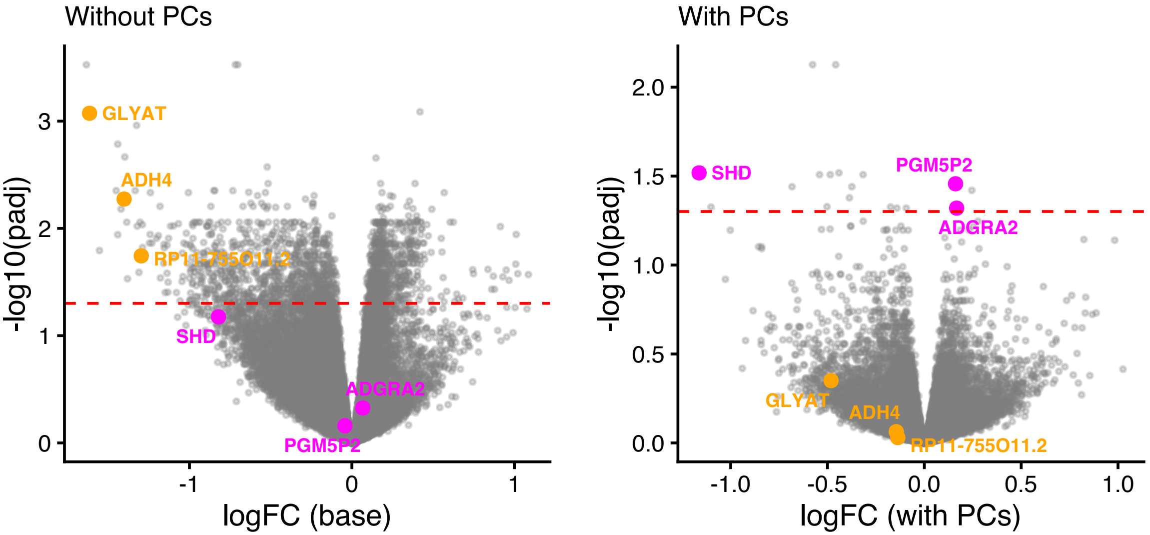
Tumour stage: t-statistic comparison



Found 6 new genes and invalidate 1,772 genes



Found 6 new genes and invalidate 1,772 genes



GLYAT: potential confounding via stromal contamination

- Liver and kidney-specific metabolic enzyme (Matsuo et al. 2012)
- Predominantly expressed in stromal cells, not tumour cells (Xia et al. 2024)
- Downregulated in liver and renal cancers

ADH4: potential confounding via hormonal regulation

- Class II alcohol dehydrogenase, primarily liver-expressed (Jelski et al. 2006)
- Regulated by steroid hormones (estrogen) (Qulali and Crabb 1992)
- Class I ADH dramatically reduced in invasive breast cancer (Triano et al. 2003)

PGM5 regulates cancer cell proliferation and invasion

- Phosphoglucomutase catalyzes interconversion of glucose-1-phosphate and glucose-6-phosphate
- Key enzyme in glucose metabolism pathway
- Downregulated in colorectal cancer tissues (Sun et al. 2019)
- Low PGM5 expression associated with poor overall survival
- Overexpression inhibits proliferation, invasion and migration
- Independent prognostic factor for overall survival

ADGRA2 linked to chemotherapy resistance

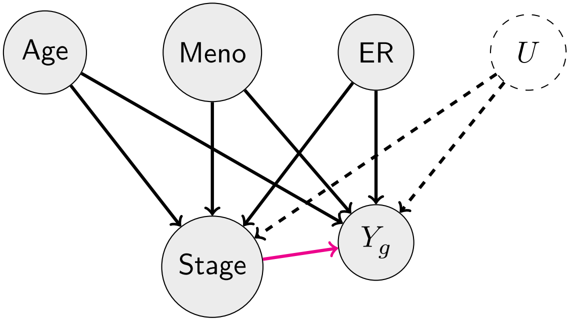
Breast Cancer Evidence:

- Amplified at 8p11.23 in NAC non-responsive tumours (Yin et al. 2024)
- Novel biomarker for chemotherapy resistance
- Associated with poor prognosis
- Overexpressed in breast cancer stem cells

General Cancer Role:

- Originally identified as tumour endothelial marker 5 (TEM5)
- Implicated in migration, proliferation, survival
- Involved in angiogenesis, invasion, metastasis
- Linked to gefitinib resistance in lung cancer

Discussion: How should we learn U?



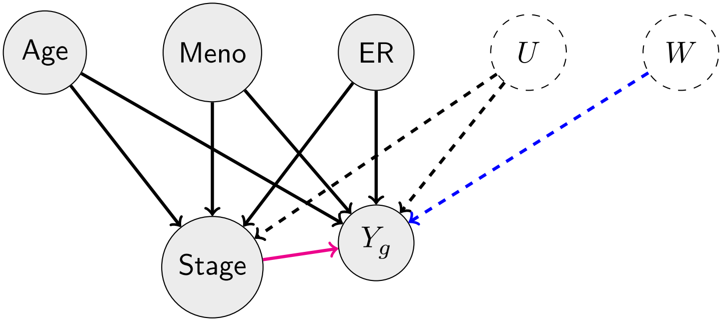
Today's lecture

1. Introduction to causal inference
2. How to incorporate causal inference in RNA-seq
3. Removing Unwanted Variation in RNA-seq

Remove Unwanted Variation (Risso et al. 2014)

- RUV uses factorization to estimate unwanted variation
- Control genes or negative control samples
- `RUVSeq` Bioconductor package:
  - RUVg: uses negative control genes
  - RUVs: uses replicate/negative control samples
  - RUVr: uses residuals from first-pass GLM

RUV: graphical model



Tumour stage effect: RUV adjustment

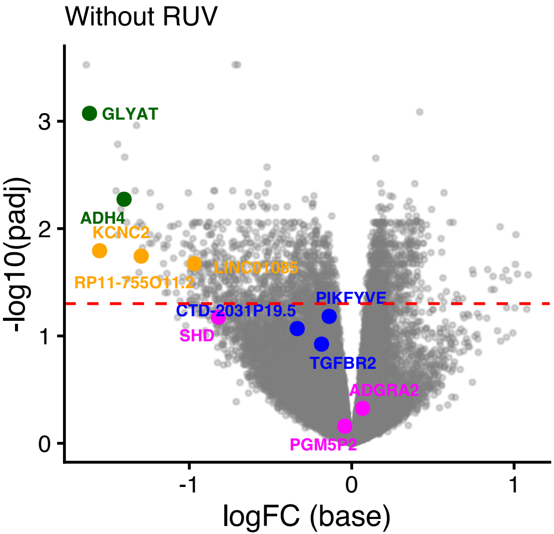
Base model:

$$Y_g \sim \text{Stage} + \text{Age} + \text{Meno} + \text{ER}$$

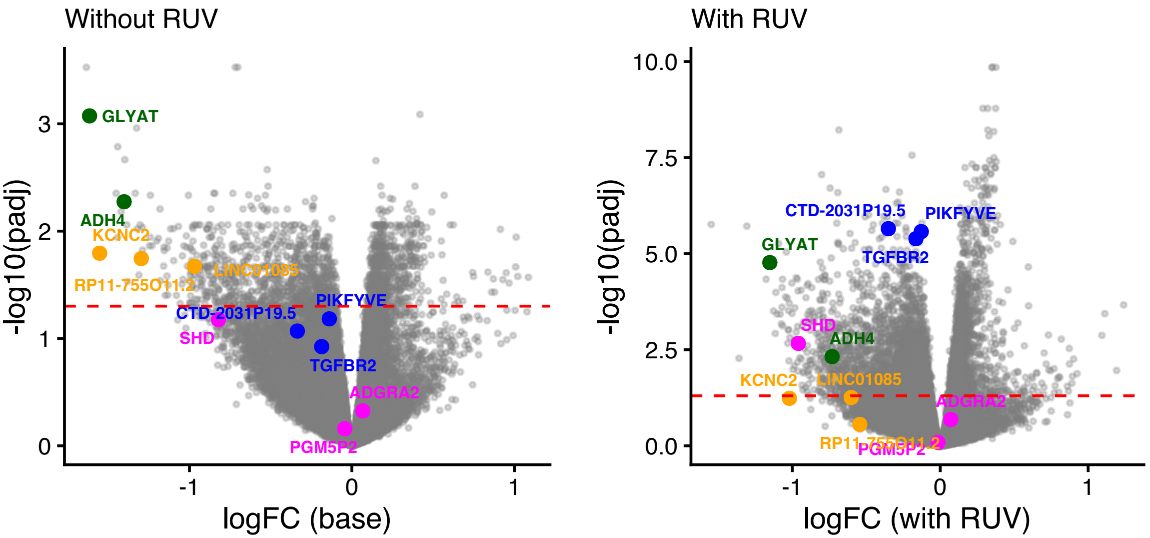
RUV-adjusted:

$$Y_g \sim \text{Stage}+\text{Age}+\text{Meno}+\text{ER}+W_1+\cdots+W_5$$

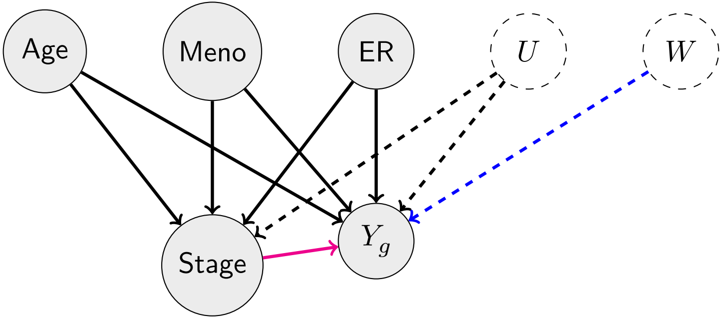
RUV adjustment: volcano plots



RUV adjustment: volcano plots



Discussion: Can RUV handle confounding effects?



Reference

Alam, Mohammad T, Hiroko Nagao-Kitamoto, Noritaka Ohga, et al. 2014. "Suprabasin as a Novel Tumor Endothelial Cell Marker." *Cancer Sci.* 105 (12): 1533–40.

Amenta, Peter S, Salim Hadad, Maria T Lee, Nicola Barnard, Deqin Li, and Jeanne C Myers. 2003. "Loss of Types XV and XIX Collagen Precedes Basement Membrane Invasion in Ductal Carcinoma of the Female Breast." *J. Pathol.* 199 (3): 298–308.

Hotelling, H. 1933. "Analysis of a Complex of Statistical Variables into Principal Components." *J. Educ. Psychol.* 24 (6): 417–41.

Reference

Jelski, W, L Chrostek, M Szmitkowski, and W Markiewicz. 2006. "The Activity of Class I, II, III and IV Alcohol Dehydrogenase Isoenzymes and Aldehyde Dehydrogenase in Breast Cancer." *Clin. Exp. Med.* 6 (2): 89–93.

Jolliffe, I T. 1986. "Principal Component Analysis and Factor Analysis." In *Principal Component Analysis*, edited by I T Jolliffe. Springer New York.

Matsuo, Moe, Kensuke Terai, Noriaki Kameda, et al. 2012. "Designation of Enzyme Activity of Glycine-N-Acyltransferase Family Genes and Depression of Glycine-N-Acyltransferase in Human Hepatocellular Carcinoma." *Biochem. Biophys. Res. Commun.* 420 (4): 901–6.

Reference

Oudart, Jean-Baptiste, Manon Doué, Alexia Vautrin, et al. 2016. "The Anti-Tumor NC1 Domain of Collagen XIX Inhibits the FAK/ PI3K/Akt/mTOR Signaling Pathway Through Av 3 Integrin Interaction." *Oncotarget* 7 (2): 1516–28.

Qulali, M, and D W Crabb. 1992. "Estradiol Regulates Class I Alcohol Dehydrogenase Gene Expression in Renal Medulla of Male Rats by a Post-Transcriptional Mechanism." *Arch. Biochem. Biophys.* 297 (2): 277–84.

Risso, Davide, John Ngai, Terence P Speed, and Sandrine Dudoit. 2014. "Normalization of RNA-seq data using factor analysis of control genes or samples." *Nat. Biotechnol.* 32 (9): 896–902.

Reference

Sun, Yifan, Haihua Long, Lin Sun, et al. 2019. "PGM5 Is a Promising Biomarker and May Predict the Prognosis of Colorectal Cancer Patients." *Cancer Cell Int.* 19 (1): 253.

Tan, Hao, Lidong Wang, and Zhen Liu. 2022. "Suprabasin: Role in Human Cancers and Other Diseases." *Mol. Biol. Rep.* 49 (2): 1453–61.

Triano, Elise A, Leslie B Slusher, Trudy A Atkins, et al. 2003. "Class I Alcohol Dehydrogenase Is Highly Expressed in Normal Human Mammary Epithelium but Not in Invasive Breast Cancer: Implications for Breast Carcinogenesis." *Cancer Res.* 63 (12): 3092–100.

Reference

Xia, Yechen, Wentao Huang, and Guang-Zhi Jin. 2024. "GLYAT Suppresses Liver Cancer and Clear Cell Renal Cell Carcinoma Progression by Downregulating ROCK1 Expression." *Transl. Cancer Res.* 13 (9): 5097–111.

Yin, Gengshen, Liyuan Liu, Ting Yu, et al. 2024. "Genomic and Transcriptomic Analysis of Breast Cancer Identifies Novel Signatures Associated with Response to Neoadjuvant Chemotherapy." *Genome Med.* 16 (1): 11.